REET NANDY

reetnandy@gmail.com | reetnandy.com | github.com/techpertz | linkedin.com/in/reetnandy | +1(518)9306116 | Willing to Relocate

Software Engineer April 2026 – Present
KeyPedia New York City, USA

As engineer #2, owned technical relations with Fortune Global 500 pharma and S&P 500 medtech customers, **discovery to production**.

- Built an MCP server that brings the entire platform into customers' AI; *a Fortune Global 500 pharma rated it deeper than its in-house agent*.
- Replaced a nightly dev-to-prod sync that was silently overwriting production with direct-to-prod writes, repointing every scraper and extraction pipeline to one source of truth, reconciling 125 tables (4.6M rows) to 0 orphans and retiring an auxiliary database.
- Root-caused an $12.6k LLM overspend to an at-least-once redelivery loop between KEDA and Celery (62k retries on 94 docs). Designed workers for fault tolerance with idempotent claims, leases, fencing tokens, attempt caps and per-service cost telemetry.
- Onboarded 11 regulatory data sources and built 3 bulk ingestion pipelines where the agents classify, extract and verify scanned PDFs on autoscaled Celery, held to ~$0.06/doc with entity deduplication and rate-limit-aware throttling.
- Raised natural-language search accuracy from 62% to 89% (95% held-out) with an eval-gated LLM parser: a 176-query benchmark, frozen held-out set, regression gates, deterministic pre-pass rules and model fallback, at 0.9s median latency.
- Refactored 10 search bars and 3 backend engines into one server-driven, permission-gated registry across 18 sources (−7k LOC).
- Built a custom BigQuery data product of 422k events, shared natively into a partner's warehouse, supporting a 2× renewal.

Founding Software Engineer May 2025 – April 2026
Sentrion AI New York City, USA

As the Startup's **first technical hire**, I single-handedly architected, developed, and deployed the company's entire technical infrastructure and product suite from the ground up and increased the MRR by 2.5x under three months. Departed on good terms when an unforeseen visa constraint prevented my STEM OPT approval, leaving only a 2-week window to transition.

- Architected data infrastructure from scratch for 60M+ rows, pairing PostgreSQL with RDS Proxy and Elasticsearch for sub-200ms search, and integrated an AWS Glue ETL pipeline to migrate 200M+ (1M+ daily) records with zero downtime.
- Built a LangGraph multi-agent system orchestrating AWS Bedrock models to automate complex workflows in the core product.
- Developed 2 FastAPI microservices and multiple RabbitMQ queues with priority routing to decouple SaaS, batch, and analytical workloads, eliminating $5k/month in vendor costs and ensuring 99.9% reliability with logging and LLM evals with LangFuse.
- Optimised cost for AWS, reducing monthly spend by $3k through autoscaling, rightsizing, and event-driven (S3) daily ingestion.
- Architected the platform's unified IAM and monetisation core, engineering a secure, tiered billing system by integrating Clerk (authentication), a custom JWT-based API credit ledger, and Stripe API.
- Served as the 1:1 technical POC for enterprises, translating business needs into engineering roadmaps to drive customer success.

PROJECTS

[AI Agent] **GraphRAG - LLM Document Compliance** (Github) *April 2025*

- Launched an Agentic SaaS with NextJS for real-time document edit, approval, and audit reports via Graph-based RAG and LLM.
- Implemented a GraphRAG and PDF parser in Python from scratch for unstructured PDFs using PDFMiner & Tesseract (OCR).
- Combined semantic chunking, NER and hybrid retrieval (vector + graph + metadata), lifting compliance accuracy to 90%.

[Python / Core] **Hierarchical Vector Database from Scratch** (Github) March 2025

- Built embedding database (library - document - chunk) with async collection mutexes; 12K ops/sec at <0.1% conflicts.
- Added 3 indexing algorithms (LinearScan/KD-Tree/LSH) for vector search on 10M vectors in 18ms.
- Led Kubernetes and Helm charts deployment along with a custom-made CLI toolkit.

SKILLS

Frontend + Backend: Python (FastAPI, Django), TypeScript (Next.js), Tailwind CSS, Node.js (Express.js)
AI / LLM: LLM evals, GraphRAG, Fine-tuning, Embeddings, Quantization, Multi-Agent, MCP and OAuth 2.1
Cloud + DevOps: AWS (EC2, Lambda, S3), GCP (GKE, Cloud Run, CDN, BigQuery), Kubernetes, Docker, KEDA, Celery, CI/CD
Data + Pipeline: PostgreSQL, MongoDB, Vector DBs, Redis, Kafka, Airflow, ETL / ELT, REST, gRPC

EDUCATION

New York University – New York City, USA September 2023 - May 2025
Master of Science, Computer Science (MS) | *Merit Scholarship Recipient* GPA: 3.7/4.0

- *Relevant Coursework*: Data Structures and Algorithms, Cloud Computing, Machine Learning, Big Data Analytics, Database Systems, Software Engineering, Object-oriented Design, Probability and Statistics
- *Graduate Teaching Assistant*: CS GY 6233 - Operating System, CSCI UA 0310 - Algorithms